Linear Regression and Logistic Regression

Abstract
Various machine learning models can be used to perform classification and predictions on datasets. In our project, linear regression is used to predict the energy efficiency of buildings based on specified features. We considered the performance of both analytical linear regression and mini-batch stochastic gradient descent based linear regression and found that the two approaches made predictions with similar accuracy. The logistic regression model is used very effectively to classify a qualitative bankruptcy dataset. The tuning of various hyperparameters in both models is also evaluated to optimize their performance.

Introduction
Logistic regression and linear regression are widely used machine learning models used in classification tasks and output prediction. These two models were analyzed for their performance and accuracy using two datasets, the Energy efficiency dataset and the Qualitative bankruptcy dataset.

The Energy Efficiency data was collected by simulating buildings using the Ecotect software to better understand the relationships between features of the building and its heating load and cooling load [1]. Calculating the heating and cooling load of a building is required to determine the specifications of the heating and cooling equipment needed to maintain comfortable indoor air conditions. Heating, ventilation and air conditioning account for much of the energy used in a building, and thus understanding the relationship between the features of a building and its heating and cooling load is key to constructing energy efficient buildings.

The Qualitative Bankruptcy data considers qualitative risk factors that were evaluated by experts through the risk estimation process and they assign appropriate levels such as positive, average and negative to these factors using their subjective knowledge and classifies each combination as bankruptcy and non-bankruptcy [2]. The prediction of bankruptcy is a critical issue because corporate bankruptcy triggers economic losses for management, stockholders, employees, customers and others. Furthermore, there are great social and economic costs to the nation.

The linear regression model effectively predicts the heating and cooling load of test data. Our experiments show that analytical linear regression and mini-batch stochastic gradient descent linear regression are equally effective for they make predictions on the test data with nearly the same error cost, which is measured as mean squared error. Tuning the gradient descent hyper-parameters had very little effect on the cost, however it did affect the convergence speed.

The logistic regression model is a reliable tool for predicting the bankruptcy of a company. Our experiments demonstrate that the size of the training dataset and the choice of mini-batch size, learning rate, epsilon, and maximum number of iterations all have a significant impact on the accuracy and cost of the model. The results of the experiments should be used to inform the selection of these parameters and align with the specific requirements of the task. An optimally configured logistic regression model can provide valuable insights and aid in making informed predictions about a company's financial stability, helping to mitigate risk and inform investment strategies.

Datasets
The energy efficiency dataset consists of 8 input variables and 2 output variables. The input variables, relative compactness, surface area, wall area, roof area, overall height, orientation, glazing area, and glazing area distribution, characterize buildings that can be simulated using specialized software. From simulations on the buildings data about the heating load and cooling load of each building which are the output variables was recorded.

Firstly, we cleaned the dataset by checking for missing values and ensuring each feature was of the correct data type. Furthermore, noticing that the scale of each variable is very different and the difference in units of the features, we decided to use mean normalization on each of the features to allow the model to compare data from different units and prevent some features from dominating. The formula for mean normalization is: $x_d^{(n)} \leftarrow \frac{x_d^{(n)} - \mu_d}{\sigma_d}$

In addition, in Tsanas et al. work with this dataset, they found that the first five input variables, relative compactness, surface area, wall area, roof area, and overall height were most strongly associated with the output variables, heating and cooling load [1]. Thus, we dropped three features, orientation, glazing area, and glazing area distribution, to improve the efficiency of our model.

The Qualitative Bankruptcy Data Set is a financial dataset designed to classify companies into two categories - bankruptcy (B) and non-bankruptcy (NB). The classification is based on seven key features, which include industrial

risk, management risk, financial flexibility, credibility, competitiveness, and operating risk. These features are described using qualitative values - positive (P), average (A), and negative (N) - which have been quantified into values 0, 1, and 2, respectively. To prepare the data for analysis, the qualitative values have been quantified and the classes have been relabeled. Specifically, the non-bankruptcy class (NB) has been designated as the negative class, with a value of 0, while the bankruptcy class (B) has been designated as the positive class with a value of 1. It is worth noting that this dataset contains no missing values, so no data cleaning was necessary.

After analyzing the second dataset, it was determined that approximately 40% of the records belong to the "bankruptcy" category. A detailed analysis was conducted on each of the following factors: "industrial risk", "management risk", "financial flexibility", "credibility", "competitiveness", and "operating risk" using the Pandas.DataFrame.describe() function. The results of this analysis are presented in the table below.

Results

We measured the performance of our analytical linear regression model on dataset 1 by calculating the minimum squared error of the trained model's prediction for both the training set and the test set. We used 80% of the data for training and 20% of the data for testing.

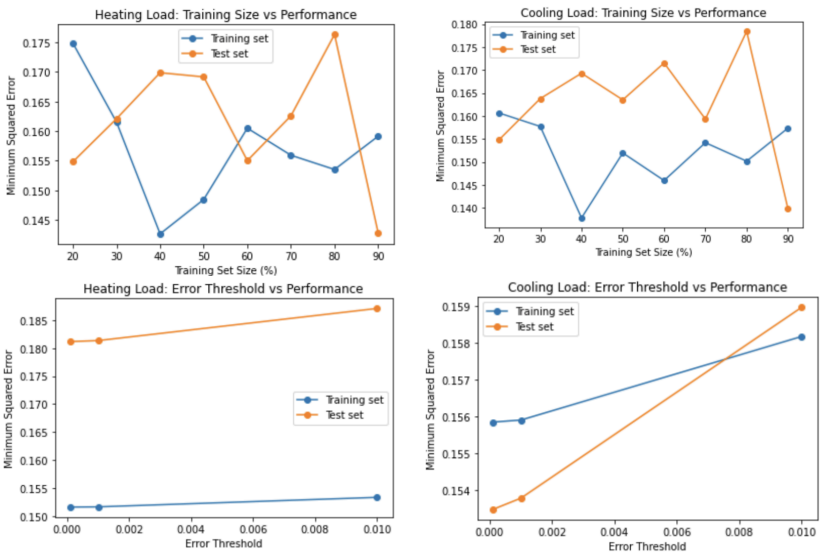
	Heating Load MSE	Cooling Load MSE
Training Dataset	0.15157106	0.15584772
Test Dataset	0.18116384	0.15347899

```
Heating Load weights
w_1 = -0.6309168635056996
w_2 = -0.3591816367543258
w_3 = 0.08307761958377266
w_4 = -0.3903748221272601
w_5 = 0.6955772228720567
b = 0.0005529986360865678

Cooling Load weights
w_1 = -0.7452234250159787
w_2 = -0.3935783865404393
w_3 = 0.01539628947644861
w_4 = -0.3912292024911996
w_5 = 0.7868914282650166
b = 0.0061054743475193254
```

Based on these weights we know that features 1 and 5, relative compactness and overall height, are most important to the model since they have the largest weights. A small change in either of these features has the most impact on the output.

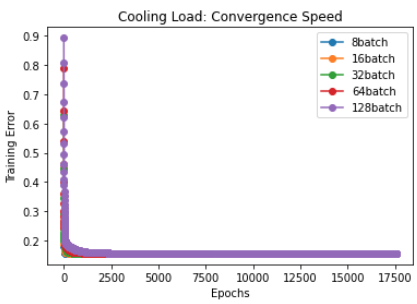
For dataset 1 when we performed analytical linear regression and experimented with the size of the training dataset, we observed that the training cost error does not significantly change or linearly decrease as we increase the size of the training set. This suggests that perhaps our model is underfitting and has a strong bias. Furthermore, it is not always the case that the training cost error is lower than the test error, nor is there an obvious trend describing this fact. Perhaps underfitting also explains this, for the randomly selected data happens to better fit the biased model.

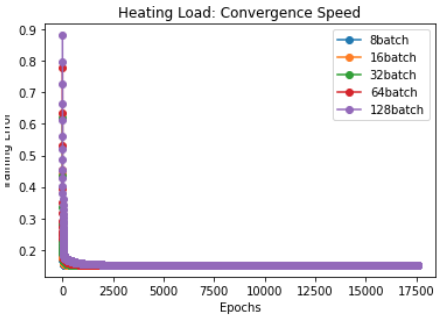


To evaluate the performance of linear regression with mini-batch gradient descent on dataset 1, various hyper-parameters were tested to understand their effects on the model's performance in terms of minimum squared error, and the models convergence rate. Firstly we experimented with the error threshold, we noticed that thresholds below 0.001 would cause the model to keep training despite having converged, and not stop until reaching the maximum epochs of 100000 regardless of all other hyper-parameters. As we increased the error threshold, the number of epochs before convergence decreased. On the other hand, increasing the error threshold decreased the performance of the model, as the error of the prediction on the test set got slightly larger. Thus, for all other experiments we set the error threshold to 0.001.

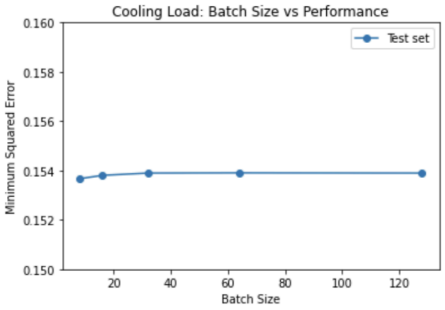
Another hyper-parameter we investigated was the batch size. Generally as the size of the batch increased the performance, in terms of error cost, decreased. However the change was marginal. Thus, we conclude that the batch

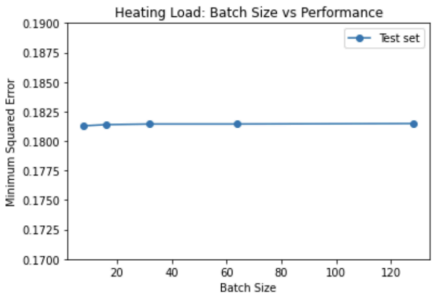
size had relatively little effect on the performance of this model on this dataset. Moreover, increasing the batch size decreases convergence speed. For convergence speed and final performance, we found that a batch size of 16 worked the best when the learning rate was fixed at 0.005, since it was significantly faster than all other batch sizes and the second smallest error.



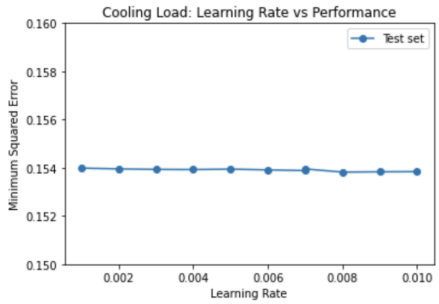


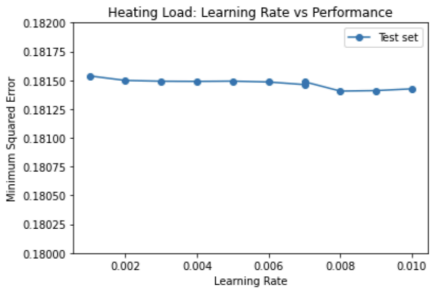
As the learning rate increased the performance of linear regression with mini-batch gradient descent was relatively unaffected by the change in learning rate and showed no linear pattern. Perhaps the changes in performance are related to noise rather than the learning rate.





Through this experimentation we can see that mini-batch stochastic gradient descent based linear regression produces approximately the same error, regardless of the combination of hyper-parameters. However, tuning the hyper-parameters can improve the convergence speed significantly.





Comparing analytical regression with mini-batch stochastic gradient descent based linear regression, you can find very little difference in terms of performance as they both produce roughly the same error of 0.181 when predicting the heating load and 0.154 for the cooling load of buildings based on the input variables. However, analytical linear regression is significantly faster than linear regression with mini-batch stochastic gradient descent. We determined this by calculating the average runtime of the fit function for both models, and using the fastest combination of hyper-parameters that we tested. We note that for larger datasets gradient descent would be faster since the time complexity of gradient descent is $O(kn^2)$ where k is the number of features and n is the total number of data points, while the time complexity of analytical solution for Linear Regression has a time complexity of $O(n^3)$. For small datasets, this difference is negligible but for large datasets the analytical solution would be unfeasible.

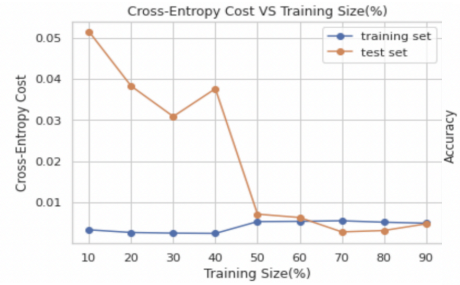
	Average Time of Model Fitting (s)
Analytical Linear Regression	0.0011013317987817572
Mini-batch SGD based Linear Regression	0.8589003793997108

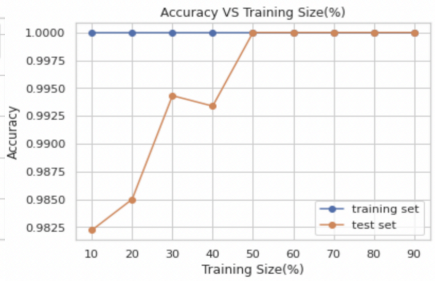
For the qualitative bankruptcy data set, the results of fully batched logistic regression with gradient descent are impressive. The accuracy of the model is 100% for both training and testing data, which is an excellent indicator of the model's ability to generalize to unseen data. Additionally, the cross-entropy cost for both the training and testing data is very low, with a value less than 0.01. This low cost error suggests that the model fits the data well and can accurately predict the class labels of the data points. The figure above also reveals that Financial Flexibility, which has the largest weight, as well as Credibility and Competitiveness, play a significant role in classifying companies into bankruptcy or non-bankruptcy.

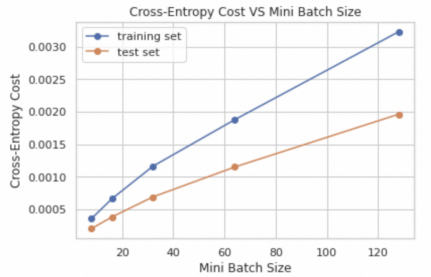
```
w1  :  -0.16367785162995058
w2  :  0.013711075148131432
w3  :  3.270511703886556
w4  :  3.085287513606897
w5  :  9.57891935300001
w6  :  0.3186102178951937
b  :  -20.154363847925016
```

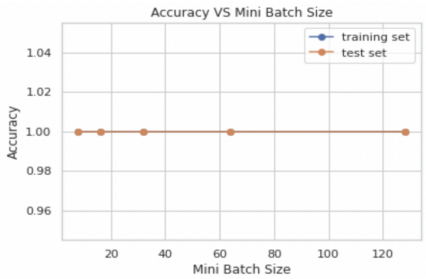
	Cross-Entropy Cost	Accuracy
Training Dataset	0.005509808152488845	1.0
Test Dataset	0.0006636566056713897	1.0

When we experimented with different sizes of training data, we noticed a clear trend in the accuracy of the test data. The accuracy increased correspondingly as the size of the training data increased. When the size of the training data reached 50% or more of the entire dataset, the accuracy of the test data reached 100%. This result suggests that larger training data sizes lead to more accurate predictions for the test data. Additionally, it was observed that the accuracy for the training data was always 100% regardless of the size of the training data. Furthermore, as the size of the training data increases, the cost error decreases dramatically, reaching less than 0.01 before the training data reaches 50% of the entire dataset. After that, the cost error continues to decrease, albeit at a slower pace, with the cost error fluctuating below 0.01 for the training data. These findings emphasize the impact that the size of training data can have on the performance of a model and the importance of careful consideration when selecting a training data size.

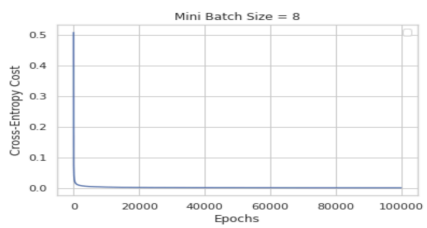


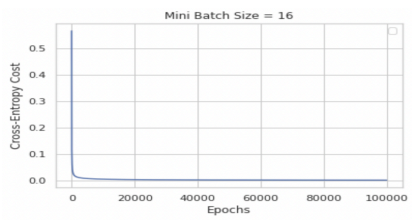


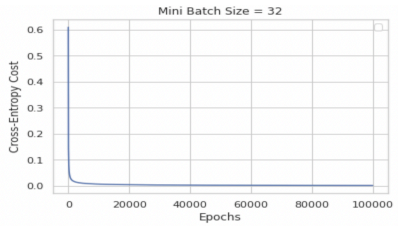


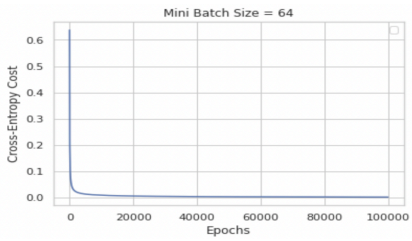


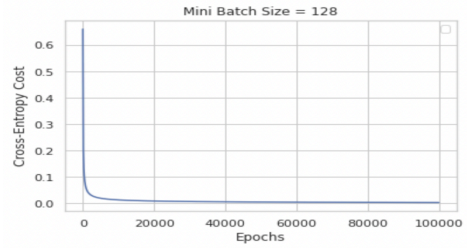
The accuracy for the training and testing datasets in logistic regression remains constant at 100% regardless of the mini-batch size. However, as the mini-batch size increases, the cross-entropy error for these datasets increases. To better understand this trend, we compared the results of a fully-batched logistic regression to those of a mini-batch method. Despite having the same accuracy of 100%, the cost error was found to be lower for the mini-batch method. This demonstrates the advantages of using mini-batch methods in logistic regression as they can produce results with a lower cost error while still maintaining high accuracy. When smaller mini-batch sizes are used, the model parameters are updated more frequently, resulting in faster convergence and a smaller cross-entropy cost.



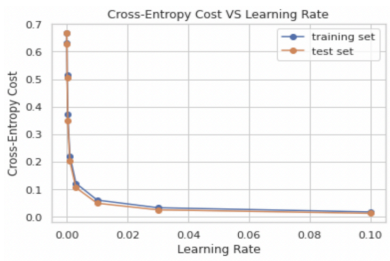


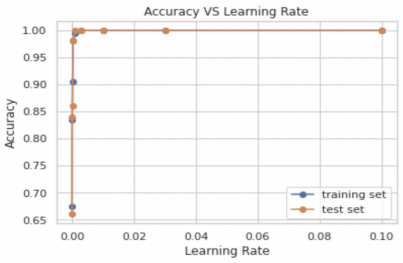


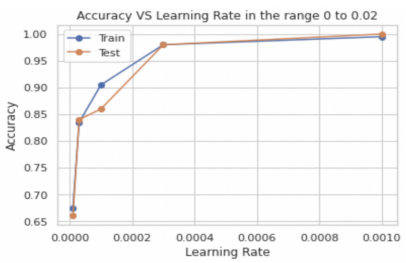




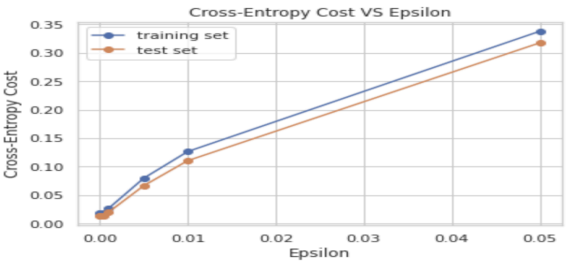
The results of our experimentation with different learning rates indicate that the cost error of both the training and testing datasets decreases dramatically with an increase in the learning rate until it reaches 0.02. Beyond this point, the cost error remains below 0.1 and remains approximately constant. Additionally, the accuracy of both the test and training datasets increases with an increase in the learning rate, with the accuracy remaining constant at 100% once the learning rate exceeds 0.001. Furthermore, when the learning rate is less than 0.001, the accuracy increases dramatically with an increase in the learning rate. These findings highlight the importance of selecting an appropriate learning rate in order to optimize the performance of a machine learning model.

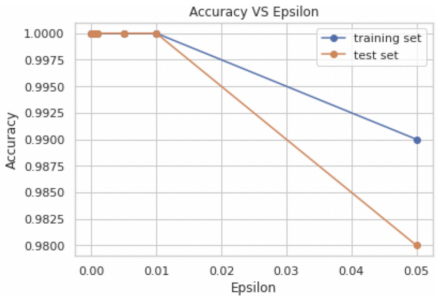






Beyond the requirements in the homework, two additional experiments were conducted to investigate the influence of different epsilon values and maximum number of iterations on the cross-entropy cost and accuracy of the model. Results showed that as the epsilon value increased, the cross-entropy cost became larger. However, when the epsilon value was small enough, it did not have any significant impact on the cross-entropy cost. With respect to accuracy, for the given dataset, the results indicated that as long as the epsilon value was not extremely large, it had no effect on the accuracy, which remained at 1.0. Regarding the maximum number of iterations, the results showed that if the value was not too small, it had no impact on the accuracy of both the training and test datasets. However, for the cross-entropy cost, as the maximum number of iterations decreased, the cost increased. Thus, the selection of epsilon and the maximum number of iterations can significantly affect the cross-entropy cost and accuracy of the model. Based on the results of the experiments, it is recommended to set the epsilon value to 0.005 and the maximum number of iterations to 50000, as there does not appear to be a further noticeable decrease in the cross-entropy cost after this point.





	Epsilon	Train Error	Test Error	Train Accuracy	Test Accuracy
0	0.00001	0.018018	0.012987	1.00	1.00
1	0.00005	0.018018	0.012987	1.00	1.00
2	0.00010	0.018018	0.012987	1.00	1.00
3	0.00050	0.018018	0.012987	1.00	1.00
4	0.00100	0.026558	0.019934	1.00	1.00
5	0.00500	0.079669	0.066551	1.00	1.00
6	0.01000	0.126547	0.110527	1.00	1.00
7	0.05000	0.337800	0.317438	0.99	0.98

	Max Iterations	Train Error	Test Error	Train Accuracy	Test Accuracy
0	100	0.515890	0.503011	0.905	0.86
1	1000	0.221133	0.202712	0.995	1.00
2	10000	0.061000	0.049629	1.000	1.00
3	50000	0.025663	0.019198	1.000	1.00
4	100000	0.018018	0.012987	1.000	1.00
5	500000	0.007024	0.004618	1.000	1.00
6	1000000	0.004781	0.003065	1.000	1.00

Discussion and Conclusion

The goal of this project is to provide analysis on Linear regression and Logistic regression on two distinct datasets. We found linear regression to be an effective model for predicting heating and cooling loads of buildings given the input provided in the energy efficiency dataset. After applying mean normalization to the dataset to prevent dominating features and to compare features of different units, we found that both analytical and mini-batch SGD based linear regression predicted the heating and cooling loads with approximately the same very low error cost. Furthermore, we found that the performance mini-batch SGD based linear regression is relatively unchanged by hyper-parameter tuning, however the convergence speed is variable. We also concluded that this model must have a relatively strong bias on this data, since there was little change in performance as we explored training set size.

Our experimentation on the Qualitative Bankruptcy Data Set using logistic regression with gradient descent showed that the model was effective in classifying companies into bankruptcy and non-bankruptcy with high accuracy and low cross-entropy cost. The results of the analysis highlighted the importance of financial flexibility, credibility, and competitiveness in the prediction of bankruptcy. The size of the training data, the use of mini-batch methods, the choice of learning rate, epsilon value, and maximum number of iterations were also found to impact the performance of the model. It is essential to consider all these factors and make informed decisions when selecting the parameters for the model. The results of the experiments should guide the selection process and align with the specific requirements of the task at hand. A well-tuned logistic regression model can provide valuable insights and help decision-makers make informed predictions about the financial stability of a company. It is a powerful tool that, when used correctly, can help mitigate risk and inform investment strategies .

Statement of Contributions

Rencheng Yu was responsible for completing all tasks in the logistic regression, while Catriona McIntosh completed all tasks in the linear regression. The report was a collaboration between Rencheng Yu and Catriona McIntosh. Farzaneh Abedini, who was a guest in the course, did not participate in any of the activities for Homework 1.

References

[1] A. Tsanas, A. Xifara: 'Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools', Energy and Buildings, Vol. 49, pp. 560-567, 2012

[2] Myoung-Jong Kim, Ingoo Han: “The discovery of experts' decision rules from qualitative bankruptcy data using genetic algorithms”, Expert Systems with Applications,Volume 25, Issue 4, pp 637-646, 2003

[3] Zhihu User. (2021, September 2). [深度学习中的epochs, batch_size, iterations详解]. Zhihu. Retrieved February 9, 2023, from https://zhuanlan.zhihu.com/p/76798271